

YIXIAN ZHANG

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EDUCATION

Tsinghua University

School of Information Science and Technology

Southeast University

Information and Computing Science

Coursework: Reinforcement Learning, Robotics, AI Agent, Optimization

August 2024 - now

Overall GPA: 3.96/4

September 2020 - June 2024

Overall GPA: 3.8/4

PUBLICATIONS (GROUPED BY RESEARCH AREA)

Sample-Efficient Reinforcement Learning

- **2025 — arXiv — First Author**

SAC Flow: Sample-Efficient Reinforcement Learning of Flow-Based Policies via Velocity-Reparameterized Sequential Modeling

Treats flow policy as a sequential model; GRU/Transformer velocity reparameterization stabilizes off-policy training and boosts sample efficiency.

- **2024 — ICLR (accepted) — First Author**

Residual Kernel Policy Network: Enhancing Stability and Robustness in RKHS-Based Reinforcement Learning

RKHS policy with residual design and representation learning for improved stability and data efficiency.

- **Zhang, Y.**, Tang, H., Wei, C., & Ding, W. (2025). Bidirectional Soft Actor-Critic: Leveraging Forward and Reverse KL Divergence for Efficient Reinforcement Learning. *arXiv preprint*. [First author]
 - **Zhang, Y.**, Tang, H., Wang, C., & Ding, W. (2025). Policy Newton Algorithm in Reproducing Kernel Hilbert Space. *arXiv preprint*. [First author]
 - Wei, C., Tang, H., **Zhang, Y.**, Wang, C., & Ding, W. (2025). Distributional Decision Transformer: Risk-Sensitive Offline RL via Quantile-Based Critics and Stochastic Return. In *IROS*. [Co-author]
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Large Multi-Agent & Networked Systems

- **2023–2025 — IEEE Trans. on Power Systems (minor revision) — First Author**

High-Dimensional Probability Preserving Scenario Reduction Using Adaptive Selection with Sampling for Large-Scale Power Systems

Scenario reduction that preserves high-dimensional probability structure for scalable multi-agent power networks.

- **2022–2023 — IEEE Trans. on Games (accepted) — First Author**

Deep Reinforcement Learning using Optimized Monte Carlo Tree Search in EWN

DRL + optimized MCTS for competitive multi-agent gameplay.

- Li, Z., Wei, Y., **Zhang, Y.**, Zhao, X., Cao, J., & Guo, J. (2024). Adaptive data processing framework for efficient short-term traffic flow prediction. *Nonlinear Dynamics*, 112(17), 15231–15249. [Co-author]

AWARDS, HONORS

Outstanding graduates in Southeast University (5%), Undergraduate Excellent Graduation Project in Southeast University (5%)

The first prize in China Undergraduate Mathematical Contest in Modeling (0.55%)